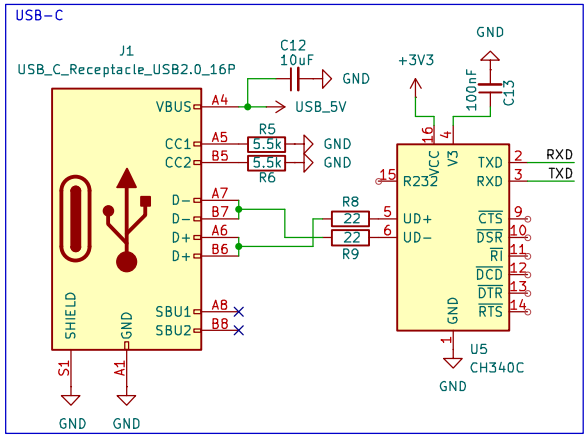
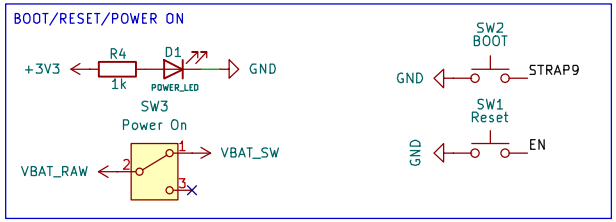
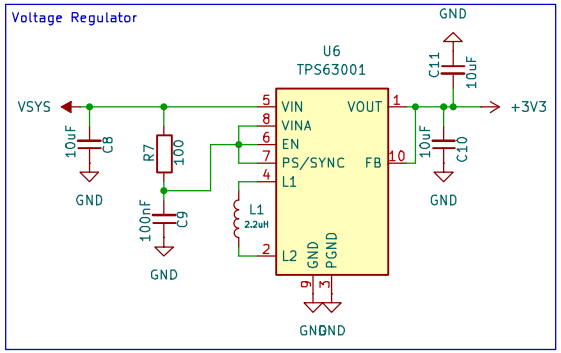
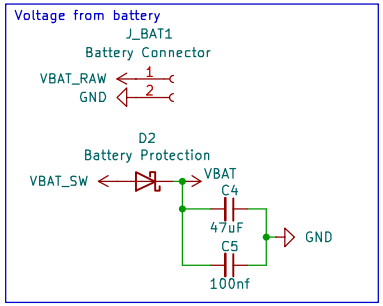
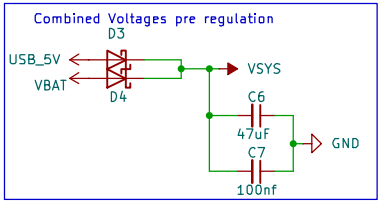
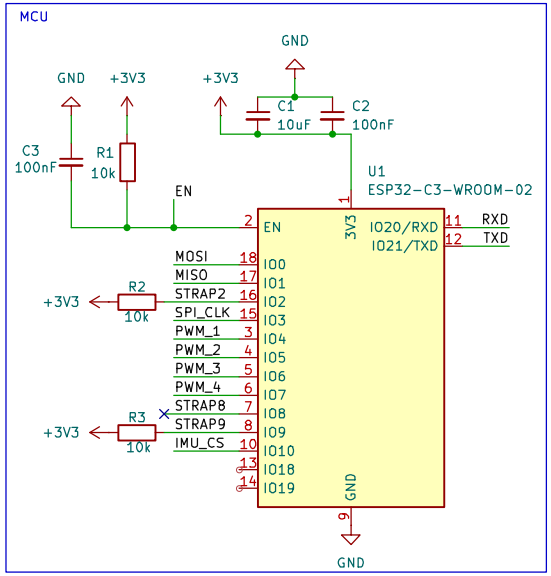

USB-C
J1
USB_C_Receptacle_USB2.0_16P
C12
10uF
GND
+3V3
100nF
C13
VBUS
A4
USB_5V
CC1
CC2
A5
B5
R5
5.5k
R6
5.5k
D-
D-
D+
D+
A7
B7
A6
B6
R8
22
R9
22
SHIELD
GND
SBU1
SBU2
A8
B8
S1
A1
R232
VCC
V3
UD+
UD-
TXD
RXD
CTS
DSR
RI
DCD
DTR
RTS
RXD
TXD
U5
CH340C

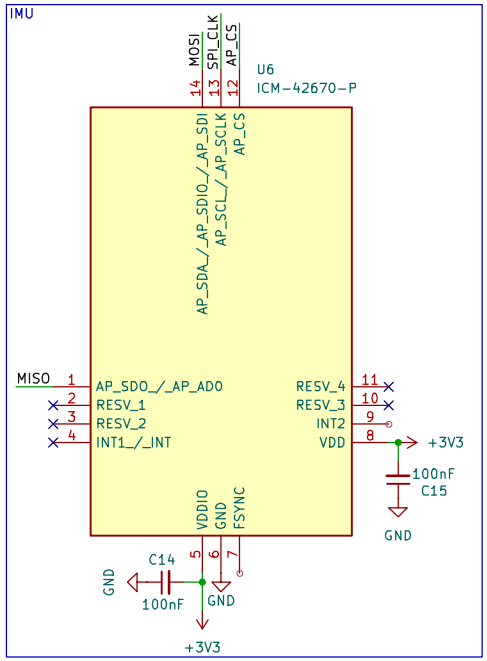

BOOT/RESET/POWER ON
+3V3
R4
1k
D1
POWER_LED
GND
SW3
Power On
VBAT_RAW
VBAT_SW
SW2
BOOT
STRAP9
SW1
Reset
EN

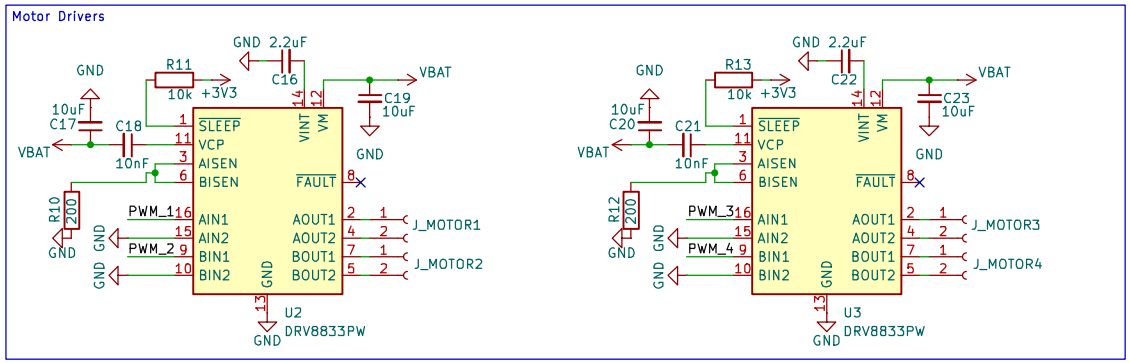

Voltage Regulator
U6
TPS63001
VSYS
VIN
VINA
EN
PS/SYNC
L1
L2
VOUT
FB
PGND
+3V3
C8
C9
C10
C11
R7
100
L1
2.2uH
GND


Voltage from battery
J_BAT1
Battery Connector
VBAT_RAW
GND
D2
Battery Protection
VBAT_SW
VBAT
C4
47uF
C5
100nf
GND


Combined Voltages pre regulation
D3
D4
USB_5V
VBAT
VSYS
C6
47uF
C7
100nf
GND


MCU
GND
+3V3
C3
100nF
R1
10k
EN
C1
10uF
C2
100nF
U1
ESP32-C3-WROOM-02
R2
10k
R3
10k
MOSI
MISO
STRAP2
SPI_CLK
PWM_1
PWM_2
PWM_3
PWM_4
STRAP8
STRAP9
IMU_CS
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO18
IO19
IO20/RXD
IO21/TXD
RXD
TXD


IMU
U6
ICM-42670-P
MOSI
SPI_CLK
AP_CS
AP_SDA_/_AP_SDIO_/_AP_SDI
AP_SCL_/_AP_SCLK
MISO
AP_SDO_/_AP_ADO
RESV_1
RESV_2
INT1_/_INT
RESV_4
RESV_3
INT2
VDD
VDDIO
FSYNC
C14
100nF
C15
100nF
+3V3
GND


Motor Drivers
GND
2.2uF
C16
R11
10k
+3V3
VBAT
C17
10uF
C18
10nF
C19
10uF
SLEEP
VCP
AISEN
BISEN
VINT
VM
FAULT
R10
200
PWM_1
PWM_2
AIN1
AIN2
BIN1
BIN2
AOUT1
AOUT2
BOUT1
BOUT2
J_MOTOR1
J_MOTOR2
U2
DRV8833PW
R13
C22
C20
C21
C23
R12
PWM_3
PWM_4
J_MOTOR3
J_MOTOR4
U3